

works best for portrait shots, which is often what people take with mobile phone cameras.

Lastly, there is matrix mode, where the camera measures the light at various points in the frame and then takes an average of them to arrive at an overall exposure for the scene. This is a very versatile option and can be used for all sorts of photos.

ISO

The ISO setting allows you to adjust the sensitivity of the sensor to light. The name is a remnant from the days of film where the ISO rating determined the sensitivity of the film to light and its speed (more sensitive film could be spun faster). Today, it is used to approximately match the sensitivity of a digital sensor to a film sensor of a particular sensitivity at a particular exposure.

Under ISO settings, you'll see values such as ISO 80, 100, 200, 400, 800, 1600, etc. The higher the number, the more sensitive the sensor becomes to light. When a sensor is more sensitive to light, you can choose to have a faster shutter speed or smaller aperture. So, for example, when you are in low light conditions, you have the option of either choosing a longer shutter speed, thereby letting the shutter stay open for longer or keep a faster shutter speed but increase the ISO value. The problem with the first option, even though it is preferable, is that you need some sort of support to stabilize the camera, such as a tripod, otherwise you risk getting a blurry image.

So then let's just crank up the ISO value, you might say. The problem with that is, when you increase the sensitivity of sensor to light, you are also boosting the amount of noise or grain in the image. The sensor has a particular sensitivity to light, that is it can output a particular signal for a particular amount of light that falls on it. By increasing the ISO, you are artificially boosting the signal it delivers, which affects the signal to noise ratio, creating more noise. The higher you push the ISO settings, the higher will be the noise in the image.

Depending upon how good the sensor is, you will get different results at a particular ISO setting for different cameras. High quality cameras will let you crank up the ISO value without adding too much noise. This is one way to judge the quality of a camera, by seeing how it performs compared to another camera at a particular ISO rating, especially at higher values.

So, which one should you select on your phone? Honestly, this is one setting that is best left at its default value, which is auto. Since most phones don't have manual shutter speed or aperture settings, there is little sense in